

Spring Math Crossword Solutions

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ONE DOWN

The area of one petal can be expressed as $2 \times \frac{1}{2} \times \int_0^{\frac{\pi}{16}} \left(\frac{8}{3} \times \cos 8\theta\right)^2 d\theta = \frac{64}{9} \times \int_0^{\frac{\pi}{16}} (\cos 8\theta)^2 d\theta$.

Substituting $\cos^2 8\theta$ for $\frac{\cos 16\theta + 1}{2}$ using the double angle identity for cosine, we get that $\frac{64}{9} \times \int_0^{\frac{\pi}{16}} (\cos 8\theta)^2 d\theta$

$$= \frac{64}{9} \times \int_0^{\frac{\pi}{16}} \left(\frac{\cos 16\theta + 1}{2}\right) d\theta$$

$$= \frac{64}{9} \times \int_0^{\frac{\pi}{16}} \left(\frac{\cos 16\theta}{2}\right) d\theta + \frac{64}{9} \times \int_0^{\frac{\pi}{16}} \left(\frac{1}{2}\right) d\theta$$

$$= \frac{64}{9} \times \left(\frac{\sin 16\theta}{32} + \frac{1}{2}\theta\right) \Big|_0^{\frac{\pi}{16}}$$

$$= \frac{64}{9} \times \frac{\pi}{32}$$

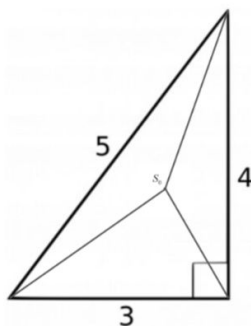
$$= \frac{2\pi}{9}$$

Since $m = 2$ and $n = 9$, $m + n = 11$.

THREE ACROSS

The optimal path of the hare consists of straight line segments joined together since the hare has to travel in straight lines if it wants to minimize the distance traveled.

The first line segment will be from the hare's starting position to one of the three vertices. The second and third line segments will be 2 of the 3 sides of the triangle.



If the endpoint of the first line segment is one of the endpoints of the hypotenuse, the hare has to travel an additional $3+4 = 7$ units along the 2 legs of the right triangle. Otherwise, the hare has to travel an additional $3+5 = 8$ units.

Let's consider a "max" point where the shortest distance the hare can travel to reach each of the corners of the field at least once is the exact same regardless of which vertex it chooses to go to first. The "max" point is where n is maximized since the hare will be closer to a vertex of the triangle if its position is anywhere else.

Let's consider this field in a coordinate plane: the bottom-left vertex is $(0,0)$; the bottom-right vertex is $(3,0)$; and the top-vertex is $(3,4)$. Since this "max" point has to be equidistant from the endpoints of the hypotenuse, it lies along the line normal to the hypotenuse, and intersecting the hypotenuse at its midpoint: $y = \frac{-3x}{4} + \frac{25}{8}$.

This "max" point also has to be one unit closer to the bottom-right vertex than the other two vertices since the hare travels an additional unit in the 2nd and 3rd segments if it decides to go through that vertex.

To find the "max" point, we set up the following equations:

$$1: x^2 + y^2 = (r + 1)^2$$

$$2: (x - 3)^2 + y^2 = r^2$$

$$3: y = \frac{-3x}{4} + \frac{25}{8}$$

Subtracting equation 2 from equation 1, we get $6x - 9 = 2r + 1 \rightarrow x = \frac{r+5}{3}$. Substituting this value of x into equation 3, we get $y = \frac{-r-5}{4} + \frac{25}{8}$.

Since this "max" point is $r + 1$ away from the bottom-left vertex, we have that $(r + 1)^2 = \left(\frac{-r-5}{4} + \frac{25}{8}\right)^2 + \left(\frac{r+5}{3}\right)^2$.

Solving the quadratic, we have only one root that produces a valid (x,y) : $r \approx 1.656$. Plugging in r , $x \approx 2.22$ and $y \approx 1.46$.

$n \approx \sqrt{2.22^2 + 1.46^2} + 3 + 4 \approx 9.66$, which rounds to 10.

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ONE ACROSS

Given a population modeled by the logistic growth curve function $P(x) = \frac{K}{(1+Ae^x)^{1/v}}$, the carrying capacity is K . Thus, the carrying capacity is **15**.

TWO DOWN

Because the variable L is normally distributed, there is approximate symmetry around the mean.

For L to round to 30, $29.5 \leq L < 30.5$. For L to round to 31, $30.5 \leq L < 31.5$. Since these two intervals are approximately symmetric around 30.5, the probability of L being in each of them is approximately the same.

For L to round to 29, $28.5 \leq L < 29.5$. For L to round to 32, $31.5 \leq L < 32.5$. Since these two intervals are approximately symmetric around 30.5, the probability of L being in each of them is approximately the same.

⋮

While the area on the right side of the mean is technically larger because it is not limited by negative values, the probability will be so small by then that it is negligible.

Because we having matching intervals for odd and even, the probability that L is odd is exactly the same as the probability that L is even - **50%**.

1. 1	2. 5
3. 1	0